COMPONENTWISE DESCRIPTION OF THE MORPHISM OF TOPOI ATTACHED TO A MORPHISM OF SIMPLICIAL SPACES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the cofinality statement ('the category (I_V^u)^opp contains as a cofinal subcategory the category of opens U with f_n^{-1}(U) superset V', source lines 425-433) is taken from the proof of the functoriality lemma (card L46171); the sheafification step uses that coverings in Y_Zar are componentwise (source lines 437-441); no verification is omitted.

Editorial note. The label lemma-describe-functoriality names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Sheaves on the simplicial site described componentwise. Same chapter, \label{lemma-describe-sheaves-simplicial-site}. Statement lines 414-417: the description in terms of which the component formulas are stated.
- The morphism of topoi f_Zar attached to a morphism of simplicial spaces. Same chapter, \label{lemma-simplicial-space-site-functorial} (card L46171). Proof lines 423-433: in its proof, for V subset Y_n open the category (I_V^u)^opp has as a cofinal subcategory the opens U subset X_n with f_n^{-1}(U) superset V, ordered by inclusion.
- Sheafification of the pullback presheaf. Sheaves, Lemma \label{sheaves-lemma-pullback-presheaves} and Section \label{sheaves-section-presheaves-functorial}. Proof lines 433-445: f_n^{-1}F_n is the sheafification of the presheaf f_{n,p}F_n.

Source reference. Original reference 09VM: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 352–352. Prerequisite pointer invoked at proof line(s) 417; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09VN: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 367–367. Prerequisite pointer invoked at proof

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line(s) 418; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 008F: sources/stacks-chow/tags.txt, source lines 315–315. Prerequisite pointer invoked at proof line(s) 438; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 008C: sources/stacks-chow/tags.txt, source lines 312–312. Prerequisite pointer invoked at proof line(s) 444; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : Y \to X$ be a morphism of simplicial spaces. In terms of the description of sheaves in Lemma 09VM the morphism f_{Zar} of Lemma 09VN can be described as follows.*

(1) *If $\mathcal{G}$ is a sheaf on Y, then $(f_{Zar,*}\mathcal{G})_n = f_{n,*}\mathcal{G}_n$.*
(2) *If $\mathcal{F}$ is a sheaf on X, then $(f_{Zar}^{-1}\mathcal{F})_n = f_n^{-1}\mathcal{F}_n$.*

Proof. The first part is immediate from the definitions. For the second part, note that in the proof of Lemma 09VN we have shown that for a $V \subset Y_n$ open the category $(\mathcal{I}_V^u)^{opp}$ contains as a cofinal subcategory the category of opens $U \subset X_n$ with $f_n^{-1}(U) \supset V$ and morphisms given by inclusions. Hence we see that the restriction of $u_p\mathcal{F}$ to opens of Y_n is the presheaf $f_{n,p}\mathcal{F}_n$ as defined in Sheaves, Lemma 008F. Since $f_{Zar}^{-1}\mathcal{F} = u_s\mathcal{F}$ is the sheafification of $u_p\mathcal{F}$ and since sheafification uses only coverings and since coverings in Y_{Zar} use only inclusions between opens on the same Y_n, the result follows from the fact that $f_n^{-1}\mathcal{F}_n$ is (correspondingly) the sheafification of $f_{n,p}\mathcal{F}_n$, see Sheaves, Section 008C. □